

POLYARNY, Russian Federation

WMO: 243220

Lat: 66.400N

Lon: 112.030E

Elev: 506

StdP: 95.39

Time Zone: 9.00 (E09)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -45.2	(c) -42.3	(d) -47.5	(e) 0.0	(f) -45.3	(g) -44.9	(h) 0.0	(i) -42.5	(j) 6.5	(k) -18.1	(l) 5.9	(m) -20.4	(n) 2.5	(o) 290	(p) 0.804

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.7	(c) 26.8	(d) 15.9	(e) 24.4	(f) 15.0	(g) 22.1	(h) 14.0	(i) 17.1	(j) 24.3	(k) 15.9	(l) 22.0	(m) 14.8	(n) 20.7	(o) 3.1	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 14.6	(b) 11.0	(c) 19.0	(d) 13.2	(e) 10.0	(f) 17.9	(g) 12.0	(h) 9.3	(i) 17.4	(j) 49.9	(k) 24.5	(l) 46.2	(m) 22.2	(n) 43.1	(o) 20.9	(p) 23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.0	6.1	5.4	-46.0	29.5	3.6	2.1	-48.6	31.0	-50.6	32.2	-52.6	33.4	-55.2	34.9
			-46.0	19.5	3.5	1.6	-48.5	20.6	-50.6	21.5	-52.6	22.4	-55.2	23.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.1	-28.7	-28.4	-17.3	-7.7	1.6	12.8	15.2	11.3	3.2	-9.4	-23.0	-28.8
	DBStd	17.42	7.93	7.07	7.80	6.66	5.81	5.32	4.78	4.42	4.85	6.62	6.99	7.71
	HDD10.0	7002	1199	1076	846	531	269	31	9	36	211	603	989	1203
	HDD18.3	9707	1457	1309	1104	781	519	177	121	221	456	861	1239	1461
	CDD10.0	377	0	0	0	0	8	116	171	77	6	0	0	0
	CDD18.3	41	0	0	0	0	0	12	24	4	0	0	0	0
	CDH23.3	305	0	0	0	0	0	103	177	25	1	0	0	0
	CDH26.7	56	0	0	0	0	0	23	31	2	0	0	0	0
Wind	WSAvg	2.9	2.8	2.6	3.0	3.4	3.4	3.3	3.0	3.0	3.0	2.9	2.5	2.4
Precipitation	PrecAvg	368	13	11	14	21	35	51	52	58	39	36	23	16
	PrecMax	522	23	22	24	42	61	90	86	132	70	52	43	54
	PrecMin	272	3	3	4	8	12	17	12	16	6	12	9	5
	PrecStd	52	4	5	6	7	14	19	18	24	16	11	8	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.2	-10.9	2.4	8.8	20.2	29.5	29.2	26.2	20.3	5.8	-1.1	-10.3
		MCWB	-8.4	-11.3	-0.7	3.0	10.0	16.5	17.4	16.4	12.5	3.5	-2.1	-10.6
	2%	DB	-13.0	-13.7	-1.0	5.4	16.2	26.1	27.1	22.9	14.2	3.1	-5.8	-13.1
		MCWB	-13.3	-14.0	-3.2	0.8	8.3	15.1	16.3	14.3	8.7	1.7	-6.1	-13.4
	5%	DB	-15.8	-15.8	-4.0	3.2	13.2	23.8	25.2	20.3	11.5	1.1	-9.8	-15.9
		MCWB	-16.0	-16.0	-5.6	-0.2	6.5	14.2	15.5	13.2	7.5	0.1	-10.1	-15.9
	10%	DB	-18.2	-18.2	-6.7	1.2	10.2	21.2	23.1	18.1	9.3	-1.0	-13.8	-18.0
		MCWB	-18.3	-18.4	-7.9	-1.6	4.8	13.0	14.8	12.5	6.2	-1.9	-14.0	-18.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.9	-11.2	-0.7	3.0	11.0	17.5	19.2	17.5	12.5	3.6	-2.2	-10.6
		MCDB	-8.8	-11.0	1.9	8.2	19.6	25.7	26.6	24.2	18.9	4.7	-1.0	-10.4
	2%	WB	-13.3	-14.0	-3.4	1.4	8.7	16.4	17.7	15.6	9.7	1.8	-6.3	-13.4
		MCDB	-13.1	-13.7	-1.2	4.5	15.6	23.7	24.9	20.5	12.6	2.8	-6.0	-13.1
	5%	WB	-15.9	-16.0	-5.5	0.1	6.9	15.1	16.5	14.2	8.0	0.1	-10.2	-15.9
		MCDB	-15.8	-15.9	-3.9	2.9	12.6	21.7	22.9	19.1	11.1	1.0	-10.0	-15.8
	10%	WB	-18.6	-18.7	-7.7	-1.4	5.0	14.0	15.3	12.9	6.6	-2.0	-13.9	-18.2
		MCDB	-18.5	-18.5	-6.6	1.2	10.0	20.5	21.4	17.5	8.7	-1.3	-13.7	-18.1
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.9	7.7	8.2	7.4	9.0	8.7	7.5	6.0	4.9	5.3	5.2
		MCDBR	7.2	7.2	8.2	8.4	10.3	11.1	10.8	10.4	8.7	5.8	9.3	6.4
		MCWBR	7.2	7.1	6.7	6.0	5.7	4.9	4.8	4.9	5.2	4.9	9.1	6.4
	5% WB	MCDBR	7.3	7.2	8.0	8.1	9.8	10.1	9.9	9.2	8.0	5.6	9.4	6.4
		MCWBR	7.2	7.1	6.7	6.3	5.8	5.2	5.3	5.2	5.3	4.8	9.2	6.4
Clear-Sky Solar Irradiance	taub	0.181	0.232	0.277	0.305	0.336	0.358	0.408	0.401	0.299	0.241	0.171	0.174	
	taud	1.863	2.055	2.087	2.115	2.239	2.406	2.247	2.212	2.485	2.196	1.888	1.874	
	Ebn at Noon	222	624	772	849	860	850	781	739	775	612	222	22	
	Edh at Noon	17	55	91	115	114	100	114	107	63	46	14	3	
All-Sky Solar Radiation	RadAvg	0.08	0.78	2.48	4.57	5.54	5.40	5.01	3.36	1.90	0.83	0.18	0.01	
	RadStd	0.01	0.05	0.13	0.23	0.34	0.50	0.40	0.43	0.21	0.10	0.02	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page